

PS 4 — The Supplier Blindspot

Domain: Trading (Wholesale Hardware Distribution)

Company: Arora Traders, Pune (industrial hardware wholesale)

Scale: 34 suppliers · ₹70–85L/month procurement · 80+ manufacturing clients

Annual Impact: 3 suppliers consistently underperform — estimated ₹8–12L annual loss unquantified

Background

Arora Traders distributes industrial hardware — bolts, fasteners, pipes, and fittings — to 80+ manufacturing clients in Pune. They source from 34 suppliers. All supplier negotiations happen based on relationship and memory. Owner Kiran Arora, 18 years in the business, carries all supplier performance data in his head.

When Kiran was hospitalised for 3 weeks last year, the business ran into 4 supplier disputes his team couldn't resolve because there was no documented performance history. Three of his 34 suppliers consistently deliver 2-5% short on quantity, charge rounded-up prices above market rates, or deliver quality that generates customer returns — but because they are 'old relationships,' they have never been challenged with data.

Problem Statement

Build an AI supplier intelligence system that:

- Builds a performance scorecard for each of the 34 suppliers from 3 years of historical transaction data
- Detects patterns: who delivers short, who delivers late, whose material generates the most customer returns
- Quantifies financial impact in rupees — not percentages — for each supplier's underperformance
- Generates a printable negotiation brief per supplier: what to push back on, with data, in a format Kiran can carry into a meeting

The Missing Attribution Challenge

In the `customer_returns.csv` file, the `supplier_id_traced` field is blank for 35% of returns. Teams must infer the likely supplier for these returns by matching the returned material type, the approximate date of return, and working backwards through goods receipts to identify which supplier's batch was most recently received. This inference is a core evaluated component — teams that skip it and only analyze the 65% with known supplier attribution will score significantly lower.

Data Files Provided

File Name	~Rows	Key Fields	Noise / Gotcha	Demo Anchor
purchase_orders.csv	~4,800	po_id, po_date, supplier_id, material_id, quantity_ordered, unit_price_quoted, delivery_promised_date	3 underperforming suppliers (VS03, VS12, VS19) have systematically higher prices built in	quantity_ordered is the ground truth — compare against goods_receipts to detect short deliveries
goods_receipts.csv	~4,800	gr_id, po_id, receipt_date, quantity_received, quality_grade, rejection_qty, invoice_amount_billed	invoice_amount_billed is based on ordered qty, not received qty — the billing-vs-received gap is the financial loss	VS03, VS12, VS19 have receipt dates 2-8 days after promised date (late delivery)
customer_returns.csv	~380	return_id, return_date, client_id, material_id, quantity_returned, reason, supplier_id_traced	35% of rows have blank supplier_id_traced — teams must infer using material + date matching	VS03, VS12, VS19 responsible for disproportionate share of returns
market_price_index.csv	~216	month, material_category, market_price_per_mt, source	Monthly benchmark prices for 6 material categories — use to flag overpricing vs market	Compare po unit_price_quoted against benchmark to detect systematic overpricing
payment_records.csv	~4,800	po_id, supplier_id, agreed_payment_days, actual_payment_date, days_late	Shows whether Arora is paying on time — context for negotiation positioning	
supplier_master.csv	34 rows	supplier_id, supplier_name, material_categories, years_of_relationship, credit_days_agreed	years_of_relationship must NOT be used as a positive performance factor — performance only	VS03, VS12, VS19 have 6, 9, and 12 year relationships respectively — old relationships, poor performance

Constraints

- Financial impact must be stated in ₹ — 'Supplier X delivered 3% short' is insufficient; it must say '₹4.2L short delivered over 3 years based on invoice vs receipt gap'
- Negotiation brief must be printable on 1-2 A4 pages — Kiran will carry it into a supplier meeting, not a laptop
- Relationship longevity (years_of_relationship) cannot be used as a positive score factor — scoring must be purely performance-based
- Must handle multi-category suppliers differently per category — VS03 may perform well on MS Pipes but poorly on GI Pipes
- Must infer supplier attribution for the 35% of returns with missing supplier_id_traced

Deliverables

- D1: Data cleaning pipeline including supplier attribution inference for missing return records
- D2: 34-supplier scorecard with multi-dimensional scoring (short delivery %, late delivery %, quality rejection %, price premium vs market)
- D3: ₹-quantified financial impact per supplier across the 3-year period
- D4: Auto-generated printable negotiation brief for the top 5 underperforming suppliers
- D5: Recommendation — which 3 suppliers should Kiran consider replacing, with suggested alternatives from the existing panel

Judging Criteria

Weight	Focus Area	What Judges Look For
40%	Financial quantification	Vague estimates ('approximately ₹X') score low. The number must be derived from: invoice_amount_billed minus (quantity_received × unit_price_quoted) for short deliveries; plus customer return value attributable to each supplier. Show the arithmetic.
35%	Missing attribution inference	Teams that analyze only the 65% of returns with known supplier attribution will be marked down. The inference method — date proximity, material matching, or probabilistic — must be documented and defensible.
25%	Negotiation brief usability	Would Kiran actually walk into a meeting with this? Specific data points, clear ask (e.g., 'We request a 3% price reduction on MS Pipes to match market rate of ₹68,000/MT'), and one-page format are evaluated.